Nathnael Dejene

Addis Ababa, Ethiopia • +251 983 568 807 • natidejene54@gmail.com
github.com/nati2126 • linkedin.com/in/nathnael-dejene-60502435a

SUMMARY

Software engineer with a background in machine learning and computer vision, focused on writing, reviewing, and evaluating high-quality Python and C++ code. Experienced in assessing model outputs for correctness, debugging complex systems, and explaining technical reasoning clearly in writing. Strong foundation in algorithms, data structures, and distributed systems, with a track record of turning research prototypes into production ready software.

TECHNICAL SKILLS

Primary languages: Python, C++

Additional languages: JavaScript, Java, SQL, MATLAB

Machine learning & AI: PyTorch, TensorFlow, deep learning, computer vision, model evaluation, quantization & pruning

Code quality: code review, debugging, algorithm analysis, data structures, clear technical writing & documentation

Backend & tools: FastAPI, Flask, REST APIs, Docker, Git, Linux

Domains: AI model evaluation & training data, backend development, computer vision, network security

EXPERIENCE

Computer Vision Engineer — Ethiopian Artificial Intelligence Institute (EAII) Jul 2025 – Present

- Develop and optimize high-performance computer vision systems in Python, contributing production-ready code to real-world deployments.
- Evaluate and debug model outputs for accuracy and correctness, identifying failure cases and improving reliability.
- Build backend services with FastAPI to support large-scale data processing and model deployment.
- Transition research prototypes into production software with cross-functional teams, documenting designs and technical decisions.

Engineering Intern — Gibe III Hydroelectric Power Station Jul 2024 – Sep 2024

- Supported maintenance and optimization of control systems for power-generation equipment, gaining hands-on experience with industrial automation and real-time monitoring.

PROJECTS

MiniTor — Onion Routing Protocol (Python) github.com/nati2126

- Designed and implemented a custom onion-routing protocol enabling anonymous communication across a network of nodes.
- Built hybrid encryption (RSA key exchange, AES data transfer) and a decentralized directory server for relay registration and client discovery.

Vision Inference Engine — ML Inference Framework github.com/nati2126

- Built a high-speed object-detection and image-classification inference framework with support for multiple deep-learning backends across diverse hardware.
- Applied quantization and pruning to reduce model size by up to 40% with minimal impact on accuracy.

EDUCATION

M.Sc. in Computer Engineering — Addis Ababa Science and Technology University 2024 – Aug 2026

B.Sc. in Electrical and Computer Engineering — Addis Ababa Science and Technology University 2020 – Jun 2025

Relevant coursework: Data Structures & Algorithms, Computer Vision, Networks, Distributed Systems, Operating Systems.